

S.3 Geography

Chapter 2: Natural Hazards

Earthquake Notes

Impacts · Measures · Level of Destruction

Textbook P.53–67

St. Margaret's Co-educational English Secondary School | S.3C

5. Impacts of Earthquake (Textbook P.53–55)

Short Term Effects	Long Term Effects
• Destruction of buildings and houses • Destruction of communication and transportation network <i>e.g. roads, railway, bridges, base stations</i> • Hinder rescue work • Destroy public utilities <i>e.g. water, gas, electricity supplies</i> • Lead to fire (leakage of gas pipes) • Tsunami • Landslides / avalanches • Flooding (dam collapses) • Land subsidence (soil → liquid → no support → collapse) • Spread of disease (dead bodies cannot be removed quickly / there is a shortage of clean water) • Aftershocks cause collapse of weakened structures (餘震) Economic loss, loss of life and property, people injured	• Obstruction of economic development • Bad effect on the psychology of the victims • Occurrence of riots and robbery (social instability)

Major Earthquakes – Data Table

Year	Location	Richter Scale	Casualties	Major Effects
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1995	Kobe, Japan	7.2	Over 6,000	Structures damaged, fires
1999	Taiwan	7.6	Nearly 2,500	Structures damaged
2004	Off the west coast of northern Sumatra, Indonesia	9.0	Over 280,000	Huge tsunami triggered and a volcano erupted
2008	Sichuan, China	7.9	Over 87,000	Structures collapsed, landslides
2011	North-east Japan	9.0	Over 20,000	Huge tsunami triggered
2015	Nepal	7.8	Nearly 9,000	Structures collapsed, landslides
2023	Turkey and Syria	7.8	Over 62,000	Structures collapsed
2024	Noto, Japan	7.5	Over 600	Structures collapsed, tsunami
2025	Myanmar and Thailand	7.7	Over 5,000	Structures collapsed
2025	Kamchatka, Russia	8.8	Less than 50	Structures collapsed, tsunami

6. Measures (Textbook P.56–59)

1. Monitoring system and warning systems *(to raise awareness and give time for early evacuation)*

- **Monitoring system:** equipment is installed to detect the signs of natural hazards
- **Warning system:** warnings are issued through media or emergency alert system when hazard strikes

Monitoring system

- Equipment to detect slight movements in the ground
- **Seismograph** is used to detect and measure seismic waves
- Scientists analyse past records of earthquakes

Warning system

- When an earthquake is detected, warnings are issued through media or emergency alert system (e.g. **J-Alert** in Japan)
- The information obtained is used to predict the **time and place of occurrence**, and the scale of possible earthquakes → evacuate or take safety precautions
- Example: **Shinkansen** in Japan conducts emergency stop when earthquake is detected

2. Hazard-proof structures *(to provide protection)*

A. For earthquake-prone regions

- **Earthquake-proof** design for buildings to prevent the collapse of structures
- Impose **laws** to ensure buildings can pass the earthquake-proof standard
- **Simple building materials** without earthquake-proof design is **prohibited**
- **Height is strictly controlled**, tall buildings are **designed to withstand strong earthquakes**

3. **Land-use planning** *(to avoid risky development)*

- Produce '**hazard-zone map**' according to level of risks (help government plan and formulate policies to minimise impact)
- Control **development** in different areas → reduce population in high-risk areas
 - → avoid building facilities with potential danger (e.g. nuclear power plants, oil)
- Establish **evacuation** routes with adequate road signs
- Large **open areas** (e.g. parks) are set up as safety evacuation areas
- School and indoor stadiums are designated as evacuation or **shelter centres**

4. **Education and drills** *(to increase public awareness, be prepared to evacuate)*

- In Japan, students are taught about the three tectonic hazards from primary school
 - → **increase public awareness** → the public knows how to **react** during a tectonic hazard
- Teach the public to pay attention to the emergency alert system
- Prepare **emergency kit**
- Set up **education centres**
- Regular large-scale **drills** → to raise **public awareness**

5. **Rescue systems** *(to reduce the number of deaths)*

- Injured people can have a higher chance to **survive**
- Well-trained **rescue team** can reduce the number of deaths
- Purchase advance equipment: **life detectors, helicopters, medicines and wireless communication networks**

6. **Post hazard financial resources** *(to rebuild the society after hazard)*

- Disaster aid
- Insurance

7. Level of Destruction Brought by Earthquakes (Textbook P.64–67)

The vulnerability of a place to tectonic hazards depends on 3 factors:

(1)	Characteristics of the affected area
(2)	Characteristics of the hazard
(3)	More Developed Countries and Less Developed Countries

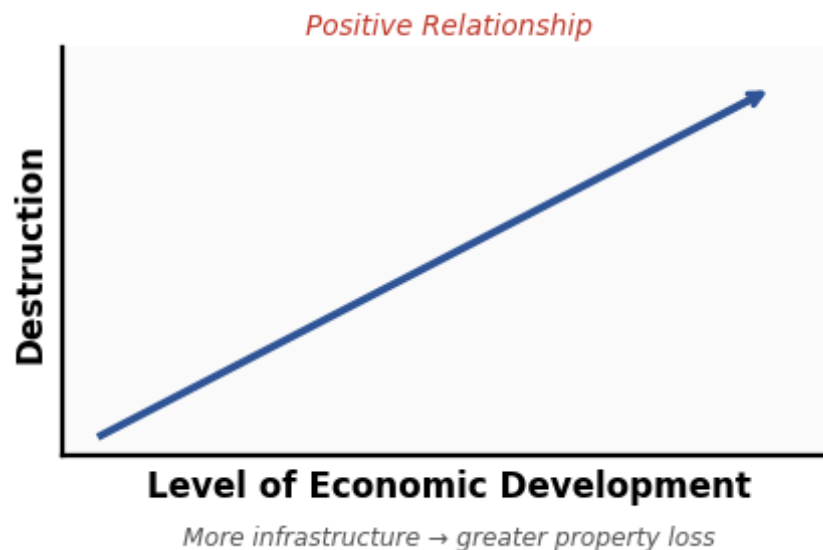
(1) Characteristics of the Affected Area

1. Population density of the area (city/town)

- The (lower / higher) the population density, the more the people exposed to the hazard

2. Level of economic development

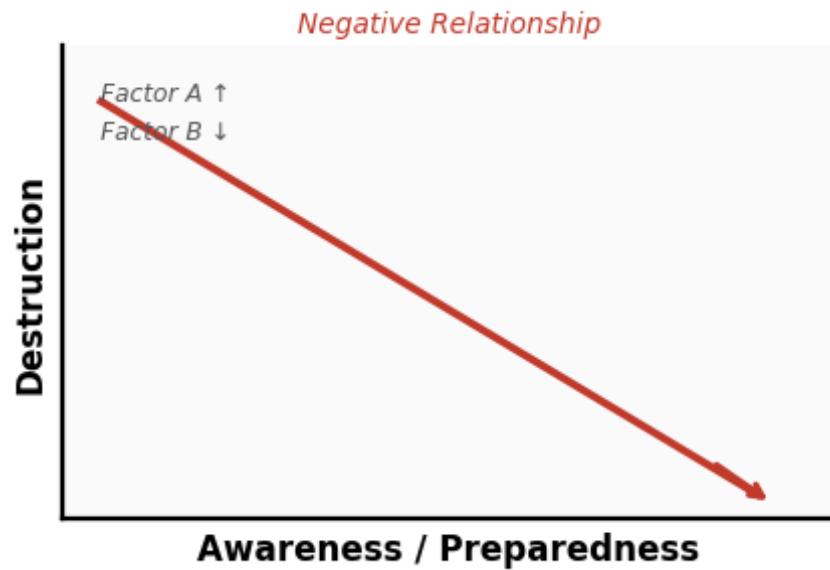
- Higher level of economic development → more economic infrastructure → greater potential loss of valuable property
 - e.g. during the night



3. Preparedness of people

- vs. low awareness → more death and casualties

- If people have **higher awareness** and are well-prepared for a hazard, they know how to react during the hazard. There is (greater / **less**) loss. e.g. evacuation



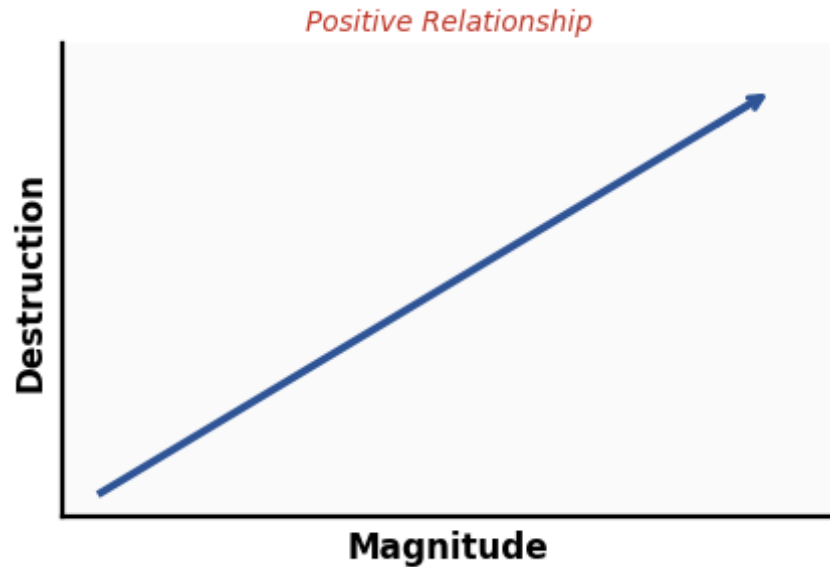
4. The weather after the hazard

- **Extreme weather** (such as cold surges and heat waves). **Increases** / Reduces death toll in areas where shelter and shade are insufficient.

(2) Characteristics of the Hazard

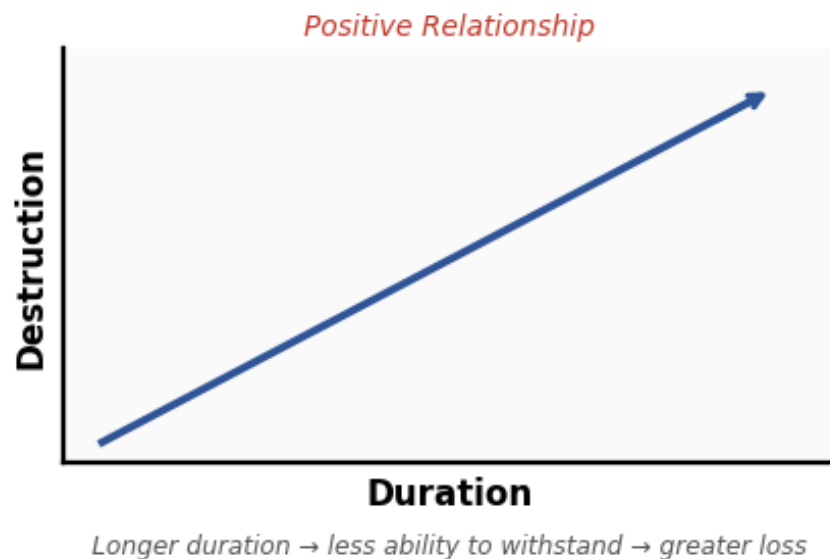
1. Magnitude

- Earthquake with a (lower / **higher**) magnitude, more energy is released and causes greater destruction.



2. Duration

- The longer the duration, the lower the ability of people and structures to withstand the hazard, results in greater losses.



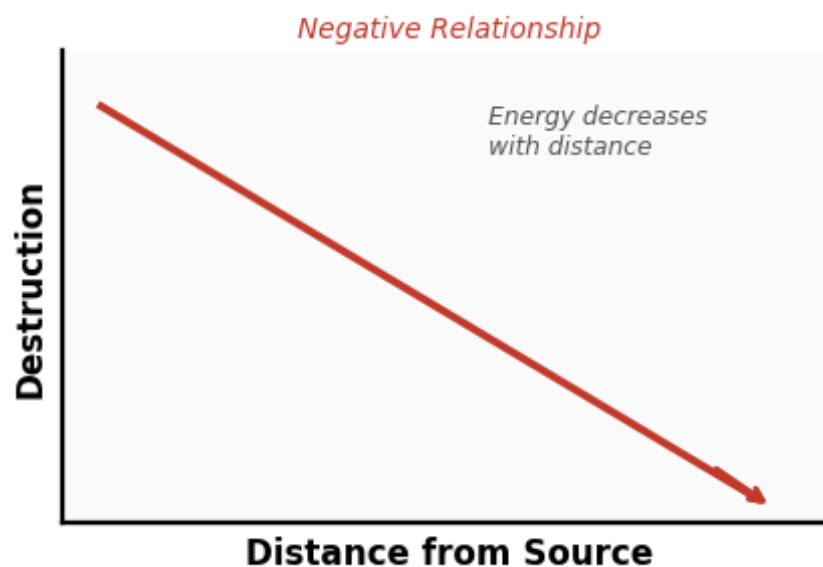
3. Time of occurrence

- USE to evacuate
- Destruction will be greater when people are not able to run away.
- e.g. at midnight (people are sleeping → low awareness) — Need to say 'Sleeping → Awareness'

4. Distance from the source of destruction

- The longer the distance from the source of destruction, the lower the destruction.

- **Energy of seismic waves** (increases / **decreases**) as it travels. Therefore, there will be more destruction when:
 - **Distance** to the epicentre (horizontal distance) is **short**
 - **Depth of focus** (vertical distance) is **shallow**



5. Frequency of the hazards

- People tend to be **less prepared** to take precautions when there are **less frequent hazards**.

6. Height of tsunami wave

- The higher the tsunami wave, the more energy is released, causing greater destruction.

(3) Level of Economic Development (MDCs and LDCs)

(refer to Classwork WS in Classwork Notes)

- In general, the **LDCs** suffer more from a hazard

1. Level of economic development (e.g. Afghanistan)

- LDCs lack **capital** and **technology** to cope with tectonic hazards

- LDCs are not able to conduct **research**, develop **warning system** and improve **rescue equipment**
- People did not receive warnings and did not evacuate, and government cannot afford **rescue** work

2. Transport network

- In LDCs, the transport network is often poorly developed → rescue work is **hindered** (not efficient) when road is blocked by landslide

3. Strength of buildings

- Many buildings in LDCs are built using **simple building materials** (e.g. stones, bricks and wood), which cannot withstand heavy shaking and fire
- **Lack of earthquake-proof designs and building standards**

4. Population density

- The population density is usually (lower / **higher**) in **cities in LDCs**, leading to higher casualties

5. Social value, education and people's awareness

- People with poor education level are (more / **less**) prepared for hazards ↗ religious beliefs
- People with strong **traditional** values and old-fashioned ideas, which is less likely to change the way they live and participate in drills and evacuation plans

6. Government efficiency

- Governments of LDCs may not respond quickly enough to help the victims
- **Corruption** prevents resources from reaching the affected area
- **Lack of law enforcement** on building codes and earthquake-proof designs